Laser Security Alarm System

Objective / Aim:

To design and implement a **Laser Security Alarm System** that detects intrusion by sensing interruption of a laser beam and activates an alarm (buzzer) as a warning signal.

Introduction / Theory:

A Laser Security Alarm is an electronic security system that uses a laser beam and a light sensor to detect unauthorized entry. The system works on the principle of **light interruption detection**.

When the laser beam continuously falls on a light sensor such as an **LDR (Light Dependent Resistor)**, the circuit remains inactive. When an intruder blocks the laser beam, the resistance of the LDR changes, causing the circuit to trigger a buzzer or alarm.

The operation of the circuit is based on:

Change in resistance of LDR with light intensity
Voltage divider principle
Switching action of a transistor

LDR Principle:

High light → Low resistance
Low light → High resistance

This change in resistance is converted into a voltage signal that controls a transistor switch.

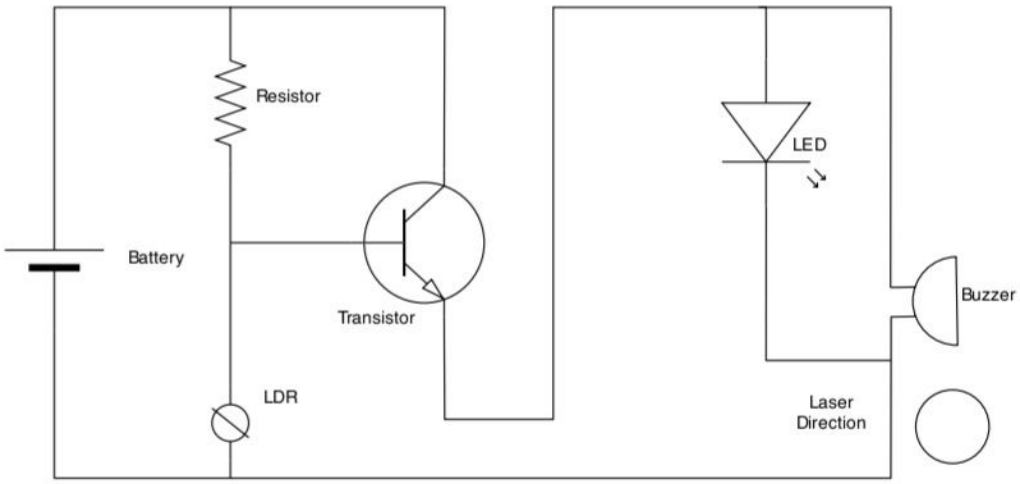
Components Required:

Laser light
LDR (Light Dependent Resistor)
NPN Transistor (BC547 or similar)
Buzzer (5V/9V)
Resistor (10kΩ)
9V Battery
Battery clip
Breadboard
Connecting wires

Circuit Diagram:

The circuit consists of:

A **laser transmitter**
A **light sensor (LDR)**
A **transistor switching circuit**
An **alarm device (buzzer)**
A **power supply**



Description of Connections:

Connect one terminal of the **LDR** to +9V supply.
Connect the other terminal of LDR to:
- A 10kΩ resistor to ground.
- Base of the NPN transistor through a 1kΩ resistor.

Connect the **Emitter** of the transistor to Ground.
Connect one terminal of the **Buzzer** to +9V.
Connect the other terminal of the buzzer to the **Collector** of the transistor.
Place the laser module so that its beam continuously falls on the LDR.

Working of the Circuit:

When the laser beam falls on the LDR:
- LDR resistance decreases.
- The base voltage of the transistor is low.
- The transistor remains OFF.
- Buzzer does NOT sound.

When the laser beam is interrupted:
- LDR resistance increases.
- Base voltage increases.
- The transistor turns ON.
- Buzzer gets supplied and sounds an alarm.

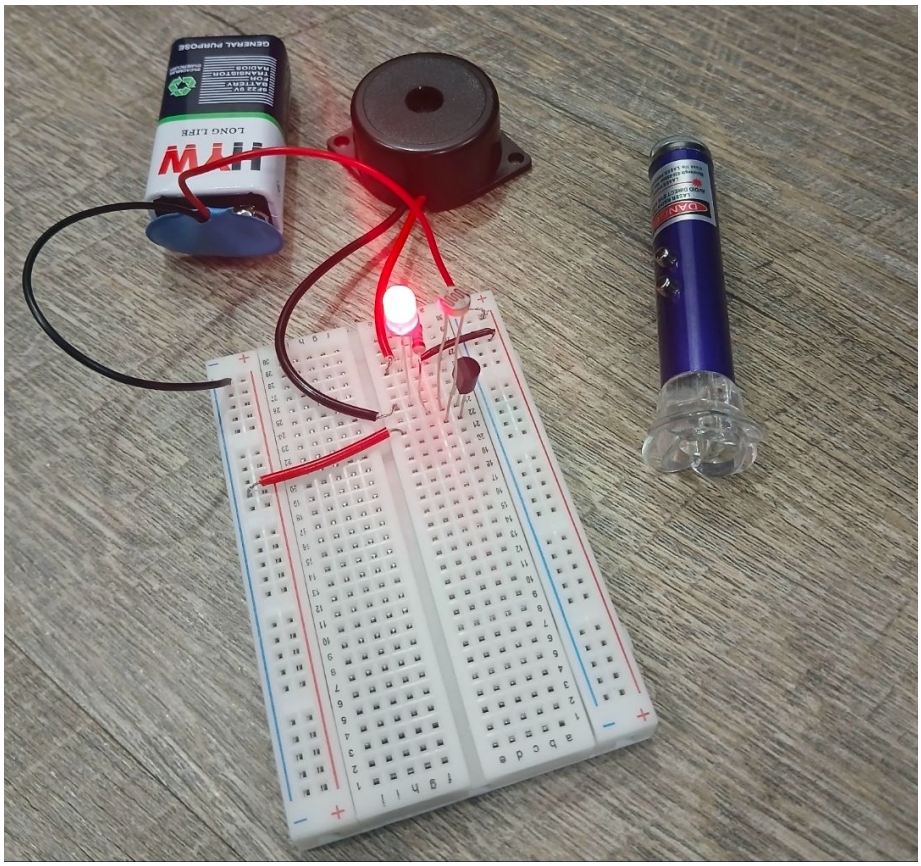
Thus, any interruption in the laser beam triggers the alarm.

Result:

The Laser Security Alarm circuit was successfully designed and tested. The buzzer sounded immediately when the laser beam was interrupted, indicating proper intrusion detection.

Applications:

Home security systems
Bank lockers
Museums
Restricted area protection
Door and window security
Military surveillance

Conclusion:

The Laser Security Alarm system effectively detects intrusions using light interruption techniques. It is a simple, economical, and efficient security solution suitable for homes, offices, and restricted areas. The experiment successfully demonstrates the practical application of LDR and transistor switching concepts.